

Ethical Hacking: The Role of Social Science in Cybersecurity

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Introduction

Ethical hacking sits at a crossroads. On one side, it's a deeply technical discipline penetration testers spend their days simulating real cyberattacks against an organization's systems, networks, and applications, probing for weak points before someone with bad intentions finds them first. On the other hand, and this is the part that often gets overlooked, the job depends on social science just as much as it depends on hacking ability. That might sound like a stretch, but isn't. Every simulation a penetration tester runs involves reading people, figuring out why employees fall for phishing emails, understanding how organizational culture creates security blind spots, and communicating findings to higher ups who may not speak a word of technical jargon. Human behavior, cultural awareness, ethical reasoning these aren't just nice additions to the skillset. They're intertwined into the work itself. This paper explores how social science research and principles shape the ethical hacking profession in practice. Drawing on concepts from our coursework, it looks at how these ideas influence the way ethical hackers operate day to day, how they engage with marginalized communities, and what kind of broader effects their work has on society as a whole.

Social Engineering and Human Behavior

If there's one area where social science and ethical hacking overlap almost completely, it's social engineering. These attacks don't target servers or firewalls, they target people. The whole idea is to manipulate someone into handing over sensitive information or doing something that opens the door for an attacker. A convincing phishing email, a fake phone call from "IT support," a well-crafted pretext none of it requires a single line of code. Just an understanding of how humans behave. Which is exactly why ethical hackers train for it. Penetration testers regularly run social engineering assessments, building realistic phishing campaigns, pretexting

scenarios, and other deceptive setups to see whether employees can spot manipulation when it's staring them in the face. Some can, and a surprising number can't. Mukherjee (2023) makes a point about this, humans are still the weakest link in any security chain, and penetration testers who ignore that reality are only doing half the job. That idea connects directly to what we've discussed in class about the psychology behind cyber threats. Cognitive biases like misplaced trust and automatic deference to authority figures are exactly what attackers prey on. A well-worded email from someone pretending to be the CEO carries more weight than it should, and attackers know it. So ethical hackers have to know it too. Studying these behavioral patterns isn't optional, it's how they build realistic simulations that actually test an organization's defenses. The end goal, as Mukherjee puts it, is helping companies develop a "human firewall." Not just better software, better instincts.

Communication and Collaboration

Nobody talks about this enough, but the hardest part of ethical hacking often isn't the hacking itself, it's what comes after. Once an assessment is finished, penetration testers have to sit down and explain what they found. And they're not just talking to other technical people. The audience might be an executive who barely knows what a firewall does, an IT team that's already defensive about their infrastructure, or a room full of employees who just want to know if they're in trouble for clicking a fake email. That range demands a completely different skill set. Translating a complex vulnerability something like a misconfigured API gateway or an unpatched authentication flaw into language that a marketing director can understand and actually act on. That's social science at work. It takes empathy, clarity, and an ability to read the room. It's been talked about in class, cybersecurity doesn't function in a vacuum. Professionals have to work across departments and build a genuine culture of security awareness, not just hand

down mandates from the IT department and hope people comply. Ethical hackers often end up right in the middle of that effort, leading or supporting training programs that teach employees how to recognize phishing attempts and social engineering tricks before they fall for them.

Conclusion

So where does all of this leave us? Ethical hacking isn't just about knowing how to break into a system. It never was. The job demands a working knowledge of human behavior, the ability to communicate clearly with everyone from a to a first-year intern to a senior level executive, a serious commitment to ethical standards, and a genuine awareness of how security decisions land on communities that are already vulnerable. Every major concept we've covered this semester social engineering, the psychology behind cyber threats, ethical responsibility, the ripple effects of cybersecurity on society maps, works directly to what penetration testers deal with in the real world. These aren't abstract ideas that live in a textbook. They show up in the field, sometimes all in the same engagement. Here's the thing that sticks with me most. Cyber threats aren't slowing down. They're getting sharper, more targeted, and harder to predict. As this happens, the social science side of ethical hacking doesn't become less relevant it becomes the whole ballgame. Knowing how to exploit a vulnerability is one thing, understanding why a person clicked the link, why a team ignored the warning, why an organization loosened security training that's where the real vulnerabilities lie.

Refernces

Allen, J. (2022, November 22). Social engineering penetration testing: Attacks, methods, & steps.

PurpleSec. <https://purplesec.us/social-engineering-penetration-testing/>

Kallberg, J., & Thuraisingham, B. (2012). Toward cyber operations: The new role of academic cyber security research and education. *Proceedings of the IEEE International Conference on*

Intelligence and Security Informatics, 132–134. <https://doi.org/10.1109/ISI.2012.6284103>

Mukherjee, A. (2023, June 11). Understanding social engineering penetration testing. *Evolve Security Automation and Orchestration by Threat Intelligence*.

<https://www.threatintelligence.com/blog/social-engineering-penetration-testing>

